

Hospital Efficiency & Patient Flow Analysis

The Project Goal

The objective was to analyze a dataset of over 9,000 patient records to determine if demographic factors (Age, Gender, Race) impact hospital wait times.

Technical Workflow

- **Advanced Data Cleaning:** Identified and corrected a recurring "stutter" typo in the gender field where entries were recorded as "Femaleemale".
- **Professional Feature Engineering:**
 - **Ordinal Encoding:** Used `OrdinalEncoder` to rank `Age Group` (Pediatric, Young Adult, Adult, Senior) so the model understands the natural progression of age.
 - **One-Hot Encoding:** Applied `OneHotEncoder` with `drop='first'` for Gender and Department to prevent mathematical redundancy.
 - **Feature Scaling:** Implemented `StandardScaler` to normalize features like `Patient Age` and `Admission Hour`, bringing them to a common scale for machine learning.

Insights

- **Operational Fairness:** The analysis revealed a surprisingly uniform system. There is no significant bias in wait times across different ages or genders as everyone waits about 35 minutes on average.
- **Staffing Bottlenecks:** While demographics don't affect the wait, **time** does. Wait times consistently peak during the **3:00 AM** shift and drop to their lowest at **Noon**.